

PROJECT DOCUMENTATION

USB PD Power Supply

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1 Introduction

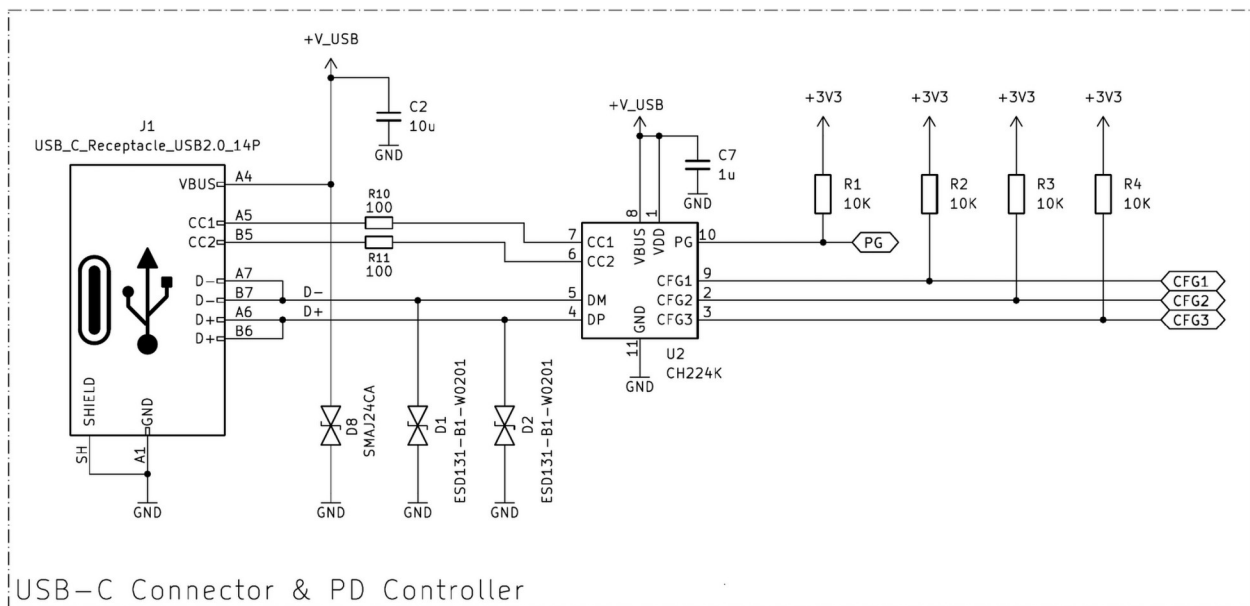
The **USB-C Power Delivery (PD)** protocol has become a standard for modern portable electronic devices. It allows the powered device to request the required voltage from a compatible power adapter.

This project focuses on designing the schematic and PCB for a device that accepts USB-C PD as its input and provides a two-screw-terminal output with manually selectable voltage using a DIP switch. The design is based on the **CH224K USB-C PD sink controller**, which communicates with the power adapter and requests the desired output voltage. This provides a simple and flexible way to obtain different supply voltages from a **USB-C PD** power source.

2 Schematic

2.1 USB-C connector & PD controller

The input stage consists of a **14-pin USB-C receptacle (J1)**. The connector is interfaced with the **CH224K USB-C Power Delivery (PD) sink controller (U2)** through the **CC and USB data lines**. The **D+ and D-** lines form a differential pair used for USB 2.0 communication, while the **CC1 and CC2 (Configuration Channel)** lines are used by the USB-C interface to detect device attachment, determine cable orientation, and perform USB Power Delivery negotiation.



The **VBUS line** provides the main input power from the USB-C adapter. The CH224K communicates with the adapter through the CC lines and requests the voltage specified by the configuration inputs. Once the requested PD voltage is negotiated, the corresponding VBUS voltage is available at the output of the input stage.

To provide protection against **electrostatic discharge (ESD) and transient events**, the VBUS, D+, and D– lines are equipped with ESD protection devices (**D8, D1, and D2**, respectively). The **100 Ω series resistors R10 and R11** are placed on the CC1 and CC2 lines to limit transient current and provide additional protection for the CH224K inputs.

The VBUS supply is locally decoupled using a **10 μ F capacitor (C2)** near the USB-C connector and a **1 μ F capacitor (C7)** placed close to the CH224K. These capacitors help reduce supply-voltage fluctuations and provide local energy storage for the controller.

The **CFG1, CFG2, and CFG3** pins of the CH224K are connected to **3.3 V through 10 k Ω pull-up resistors (R2, R3, and R4)**. These configuration inputs determine the voltage requested from the USB-PD power adapter. In this design, the configuration signals are connected to the external voltage-selection circuitry, allowing the desired output voltage to be selected using a DIP switch.

The **PG (Power Good)** output is pulled up to 3.3 V through **R1 (10 k Ω)**. This signal provides an indication of whether the PD negotiation has successfully completed and a valid power source is available.

2.2 Voltage selector switch

The **CFG1, CFG2, and CFG3** pins of the CH224K are connected to a **6-position DIP switch (SW1)**, which allows the user to manually select the desired output voltage. Each CFG pin can be either connected to **GND** through the DIP switch or left disconnected. When a switch is open, the corresponding **10 k Ω pull-up resistor** holds the CFG pin at 3.3 V, representing a logic **1**. When the switch is closed, the pin is connected to GND, representing a logic **0**.

By combining the pull-up resistors with the DIP switch, different logic combinations can be selected manually. Each combination corresponds to a specific USB-PD voltage requested by the CH224K from the connected power adapter.

The available voltage-selection combinations are shown in the table next to the schematic. A **1** indicates that the corresponding CFG pin is pulled up to 3.3 V, while a **0** indicates that the pin is pulled down to GND. The **X** denotes a *don't-care* condition, meaning that the state of that pin does not affect the selected output voltage.

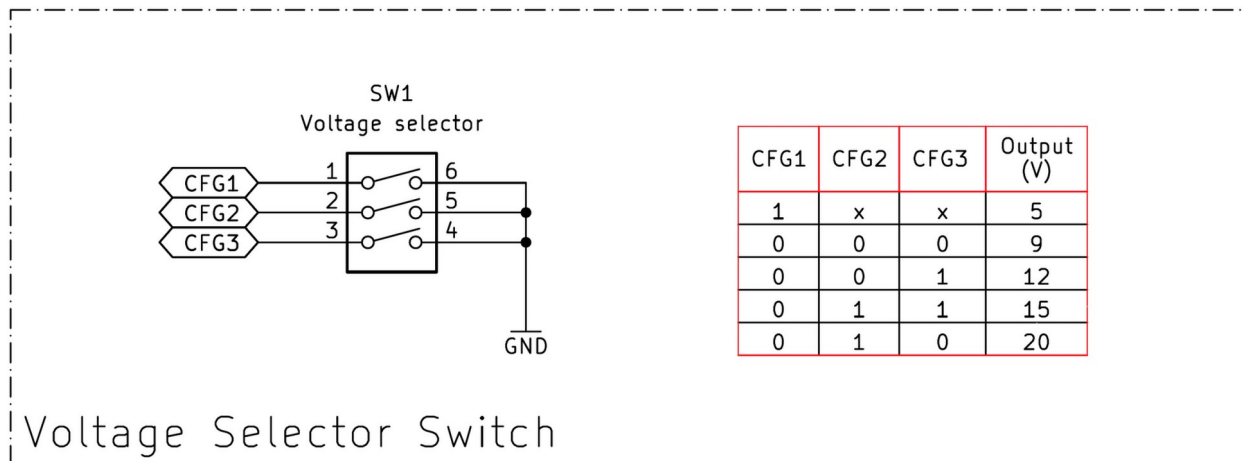


Figure 2: Voltage selector switch

2.2 3.3 V Regulator

A **3.3 V supply** is required to provide the logic-high level for the **CFG and PG pins** of the CH224K and to power the **multiplexer used for the output-voltage indication LEDs**.

The input voltage, **VBUS (+V_USB)**, is supplied to the **TPS70933DBVR low-dropout regulator (LDO)**. The regulator accepts an input voltage between **2.7 V and 30 V** and provides a fixed **3.3 V output** in this design. This makes it suitable for generating the low-voltage supply required by the control and indication circuitry across the available USB-PD output-voltage range.

The **input capacitor C1** and **output capacitors C4, C5, and C6** provide local decoupling and help maintain a stable regulator output by reducing voltage fluctuations and transient disturbances.

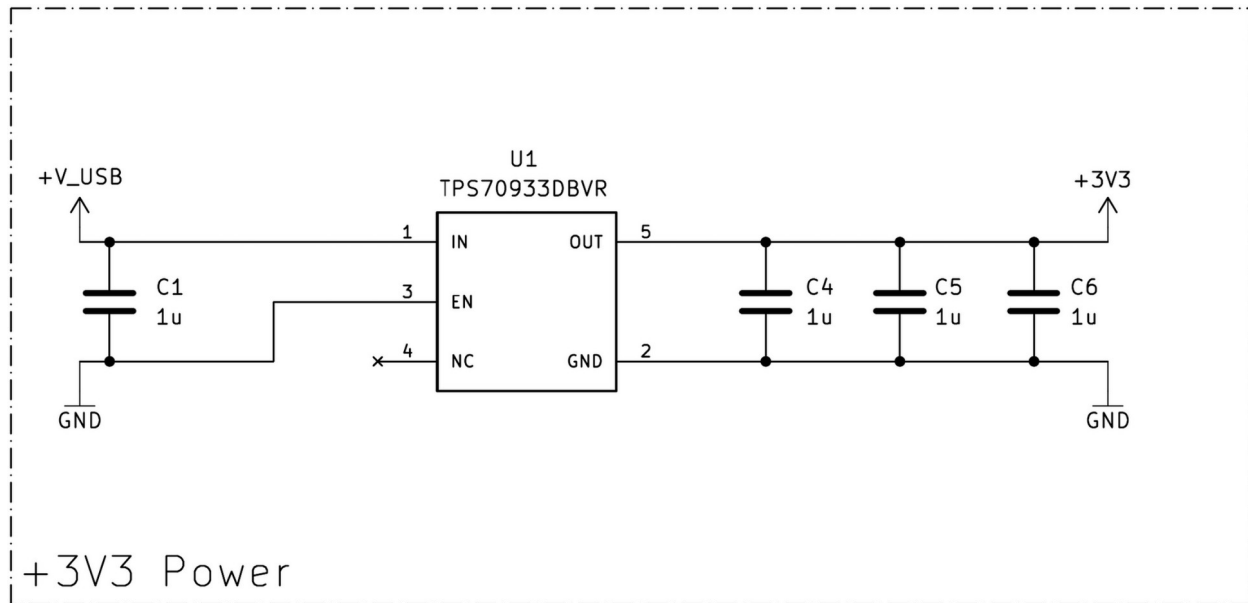


Figure 3: 3.3 V Regulator

2.3 Error LED

The **error LED** is controlled by the **PG (Power Good)** output of the CH224K through an **N-channel MOSFET (Q1)**. The PG signal indicates whether the USB-PD negotiation has been successfully completed and a valid power output is available.

The error LED is **turned on when the PD negotiation fails**, such as when the connected power adapter does not support the selected output voltage or does not support the USB Power Delivery protocol. When the negotiation is successful and the requested voltage is available, the PG signal changes state and the error LED is turned **off**.

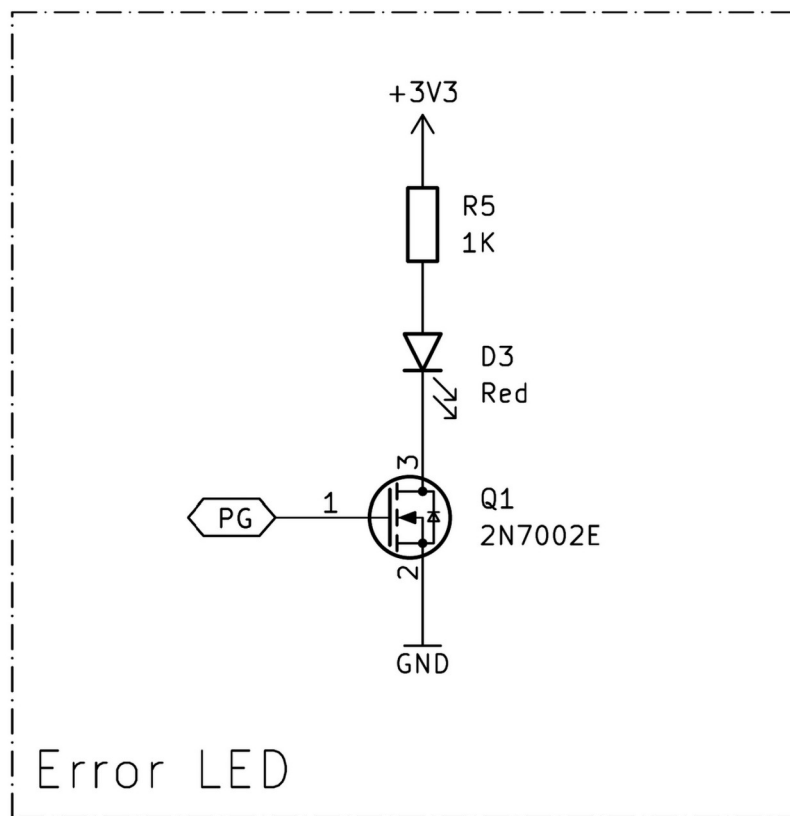


Figure 4: Error LED

2.4 LED voltage indicator

The selected output voltage is indicated using a set of LEDs controlled by the **MC14051BDR2G multiplexer (U3)**. The logic states of the **CFG1, CFG2, and CFG3** pins, determined by the voltage-selection DIP switch, are used to select the corresponding multiplexer channel.

Based on the selected CFG combination, the multiplexer activates the **LED corresponding to the requested output voltage**, providing a visual indication of the selected voltage. This allows the user to easily identify the voltage requested from the USB-PD power adapter.

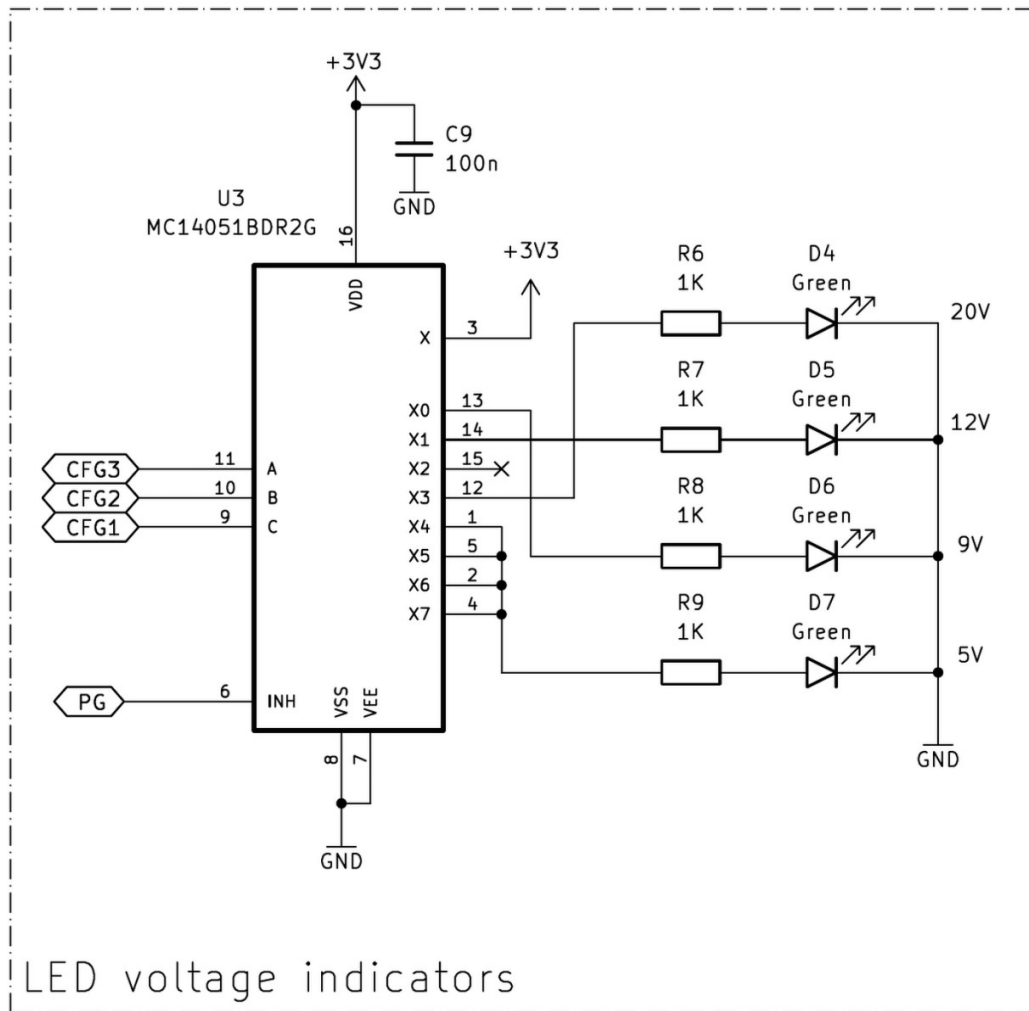


Figure 5: LED voltage indicator

2.4 Output terminals

Output is protected from inverse polarity via P-Channel MOSFET (**Q2**). Zener diode (**D9**) and resistor (**R12**) are protecting the PMOS gate from inverse voltage that is higher than rated V_{gs} . Output is decoupled by 10 uF capacitor

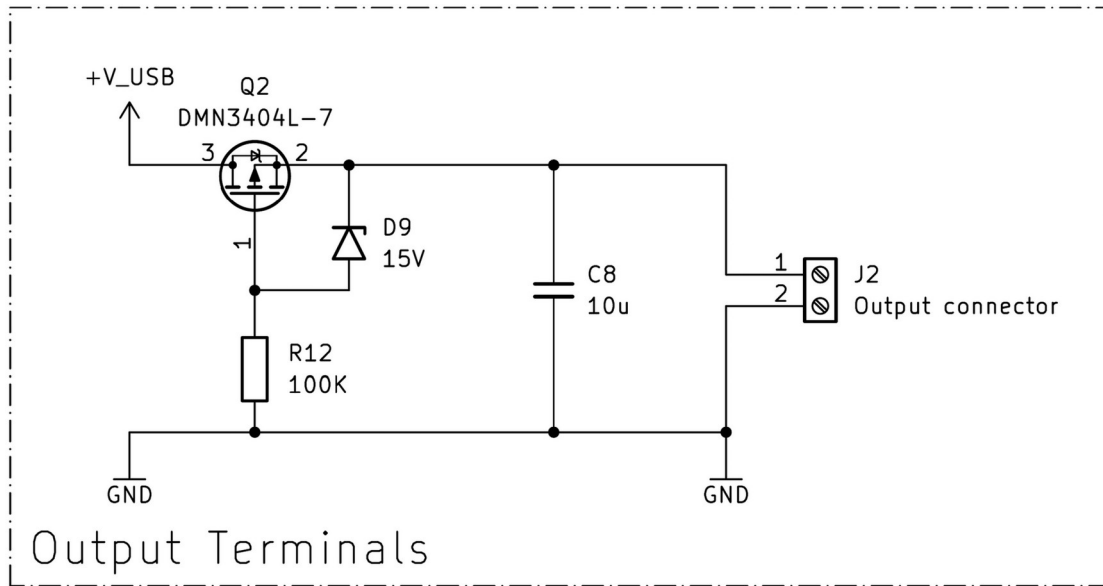


Figure 6: Output terminals

3 PCB

The PCB for this project consists of **four copper layers**. The two inner layers are primarily used as **ground planes**, while the top and bottom layers are used for component placement and signal routing. Different trace widths are used depending on the type of connection. **10 mil (0.254 mm)** traces are used for general signal routing, while **20 mil (0.508 mm)** traces are used for power connections.

The USB **D+ and D-** lines form a differential pair and require controlled impedance to ensure reliable USB 2.0 communication. The differential pair on this PCB is designed for a **90 Ω differential impedance**, which is the nominal target for USB 2.0. The spacing between the D+ and D- traces is **8 mil (0.203 mm)**. Based on the PCB stack-up and the differential impedance calculator, a trace width of **11.44 mil (0.291 mm)** is required to achieve the desired 90 Ω differential impedance.

Since the CH224K supports USB-PD power delivery of up to **100 W**, a 20 mil trace width may not provide sufficient current-carrying capacity for the VBUS path, particularly at higher power levels. To address this, **copper-filled zones (power planes)** are used for the **VBUS and 3.3 V power rails**. These larger copper areas provide a significantly greater conductive cross-sectional area than narrow traces, reducing resistance and voltage drop while allowing higher currents to be carried safely.

The use of dedicated power planes also helps distribute heat over a larger area and reduces the risk of excessive temperature rise in high-current sections of the PCB.

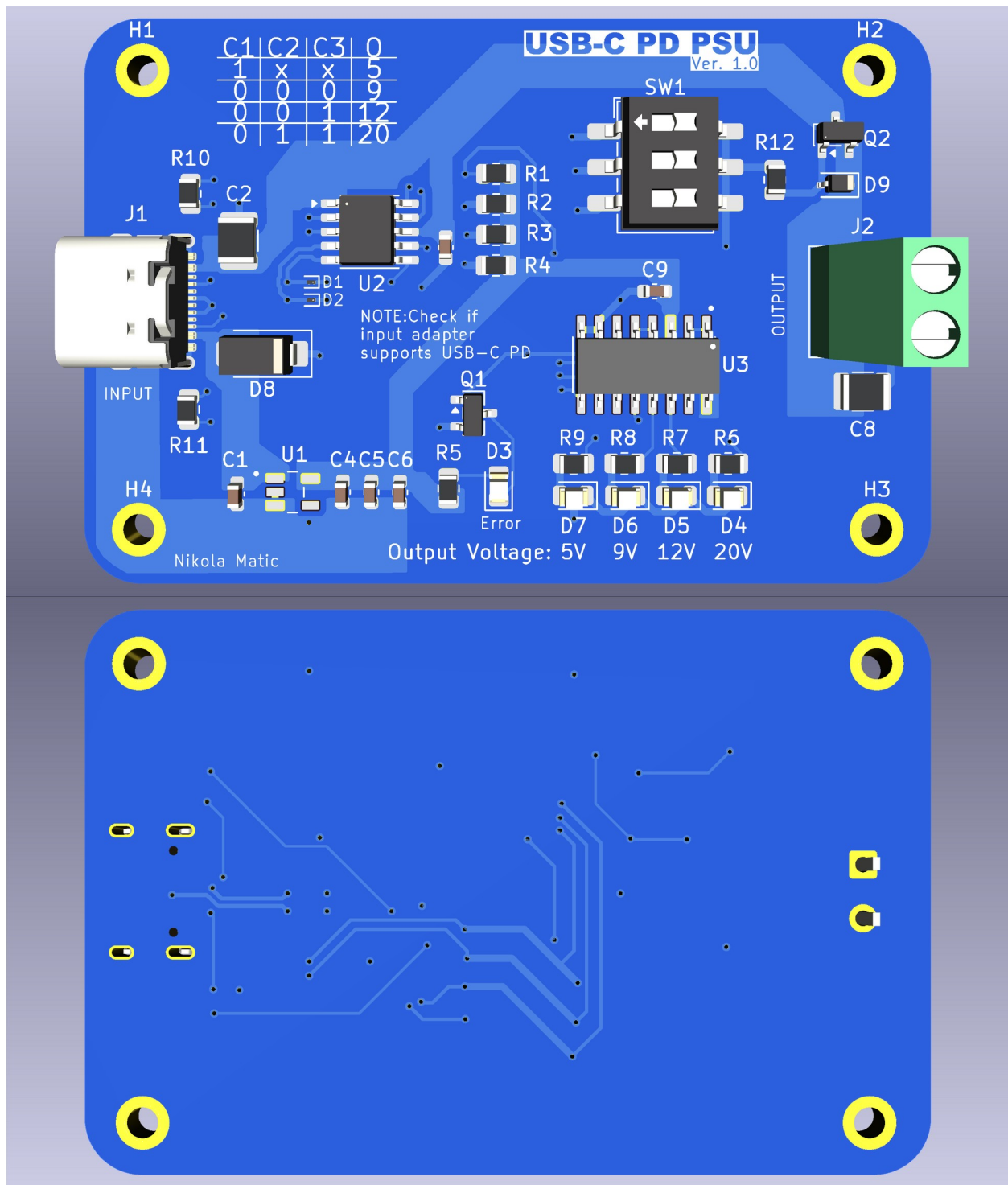


Figure 7: 3D view of the PCB (front & back)

4 BOM

Reference	Qty	Value	Mfr. No.	Price per piece (€)	Price (€)
C1,C4,C5,C6,C7	5	1u	CC0603KPX7R9BB105	0.91	4.55
C2,C8	2	10u	C1210C106K5PAC	1.7	3.4
C9	1	100n	CC0603MRY5V9BB104	0.09	0.09
D1,D2	2	ESD131-B1-W0201	ESD131B1W0201E6327XTSA1	0.24	0.48
D3	1	Red	Q62702P5178	0.26	0.26
D4,D5,D6,D7	4	Green	Q62702P5179	0.26	1.04
D8	1	SMAJ24CA	SMAJ24CA	0.53	0.53
D9	1	15V	BZX384C15-HE3_A08	0.41	0.41
J1	1	USB_C_Receptacle_USB2.0_14P	UJ20-C-G-MSMT-TR-68	2.39	2.39
J2	1	Output connector	1985807	0.96	0.96
Q1	1	2N7002E	2N7002ET1G	0.2	0.2
Q2	1	DMN3404L-7	DMN3404L-7	0.35	0.35
R1,R2,R3,R4	4	10K	CR0805-JW-103ELF	0.09	0.36
R5,R6,R7,R8,R9	5	1K	CR0805-JW-102ELF	0.1	0.5
R10,R11	2	100	CR0805-JW-101ELF	0.1	0.2
R12	1	100K	ESR10EZPJ101	0.13	0.13
SW1	1	Voltage selector	219-3LPS	0.69	0.69
U1	1	TPS70933DBVR	TPS70933DBVR	1.3	1.3
U2	1	CH224K	CH224K	0.22	0.22
U3	1	MC14051BDR2G	MC14051BDR2G	0.68	0.68
				Total:	18.74